

CIVL4333 Advanced Concrete Structures

Prestressed Flexural Members

Problem 1 – Elastic stress distribution

Calculate the elastic stress distribution for the beam shown below.

- The applied prestressing force on the beam, $P = 2400 \text{ kN}$
- The uniformly distributed load including self-weight = 30 kN/m
- The area of beam cross section, $A = 320,375 \text{ mm}^2$
- The second moment of inertia, $I_g = 5.02 \times 10^{10} \text{ mm}^4$

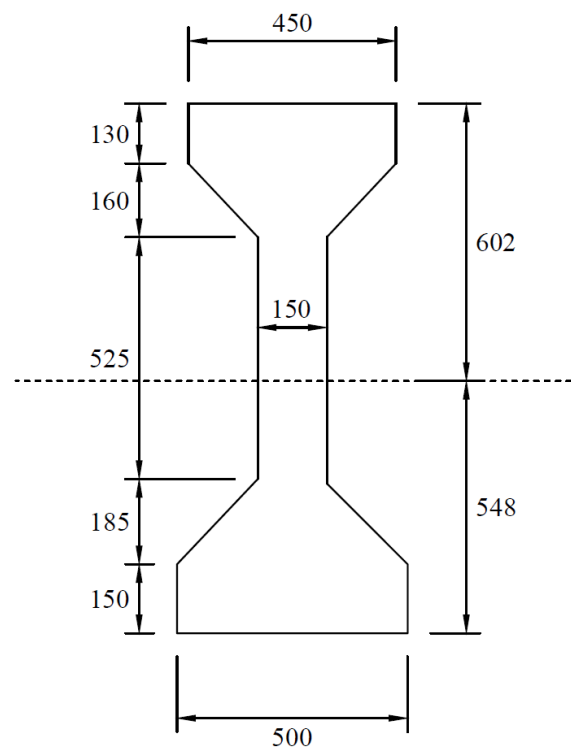


Figure 1.1 - Beam cross-section

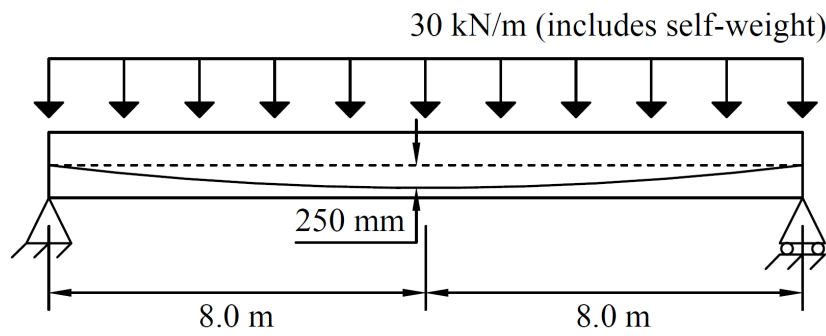


Figure 1.2 – Beam elevation



Solution

Calculate the extreme fibre stresses at midspan due to prestressing and applied loads:

$$P = 2400 \text{ kN}$$

$$P.e = 2000 \times 250 \times 10^{-3} \text{ kNm}$$

$$M = \frac{wl^2}{8} = \frac{30 \times 16^2}{8} = 960 \text{ kNm}$$

Stresses due to P (compressive) =

$$\frac{P}{A_g} = \frac{2400 \times 10^3}{320,375} = 7.5 \text{ MPa (compression)}$$

$$\sigma_t = \sigma_b = 7.5 \text{ MPa}$$

Assume tension at top and compression at bottom. Stresses due to P.e:

$$\sigma_t = \frac{P.e.y_t}{I_g} = \frac{2400 \times 10^3 \times 250 \times 602}{5.02 \times 10^{10}} = 7.20 \text{ MPa (tension)}$$

$$\sigma_b = \frac{P.e.y_b}{I_g} = \frac{2400 \times 10^3 \times 250 \times 548}{5.02 \times 10^{10}} = 6.55 \text{ MPa (compression)}$$

Stresses due to M:

$$\sigma_t = \frac{M.y_t}{I_g} = \frac{960 \times 10^6 \times 602}{5.02 \times 10^{10}} = 11.51 \text{ MPa (compression)}$$

$$\sigma_b = \frac{M.y_b}{I_g} = \frac{960 \times 10^6 \times 548}{5.02 \times 10^{10}} = 3.57 \text{ MPa (compression)}$$

Combine all the stresses:

$$\sum \sigma_t = 7.5 - 7.20 + 11.51 = 11.81 \text{ MPa (compression)}$$

$$\sum \sigma_b = 7.5 + 6.55 - 11.51 = 3.57 \text{ MPa (compression)}$$

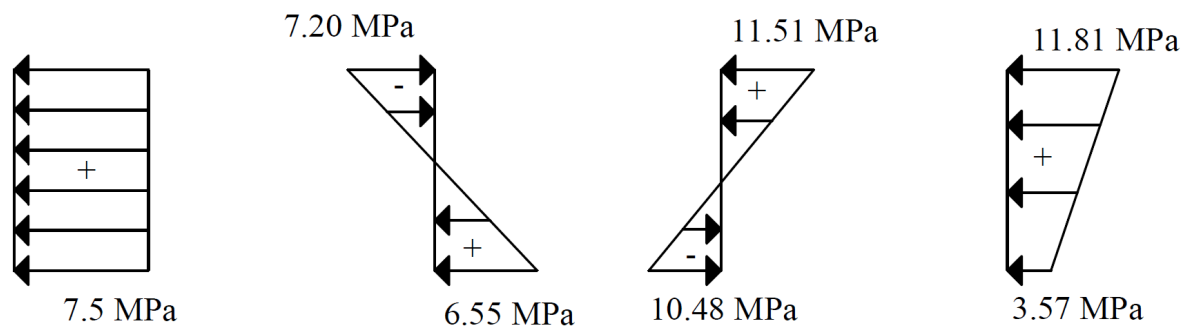


Figure 1.3 – Extreme fibre stresses due to prestressing and applied loads



Problem 2 – Calculation of stresses in an uncracked prestressed beam

The cross-section of a post-tensioned concrete beam spanning 15 m, simply supported, is shown in Figure 2.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 300 mm. The tendon comprises of 8 number 12.7mm strands. The tendon is stressed to 80% of the breaking load. Properties of the materials are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

1. Determine the stresses in the beam section at midspan due to prestress acting alone
2. Determine the decompression moment at midspan
3. Determine the stresses at a midspan due to prestress, self-weight and a uniform live load of $18 \text{ kN} / \text{m}$.
4. Is the beam cracked under the above loading?

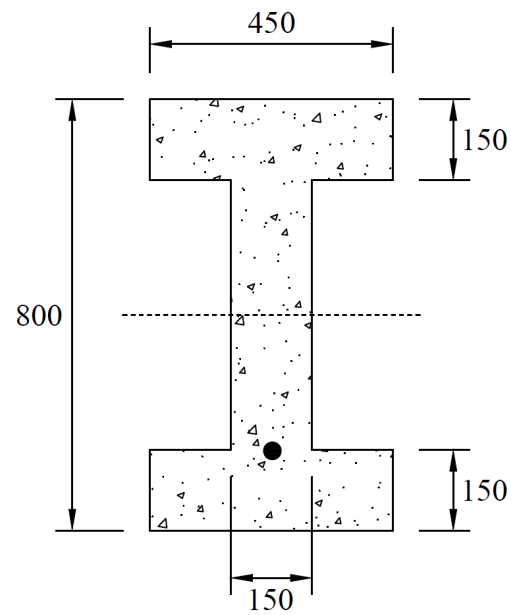


Figure 2.1 - Uncracked prestressed beam section

Solution

Calculation of stresses in an uncracked prestressed beam:

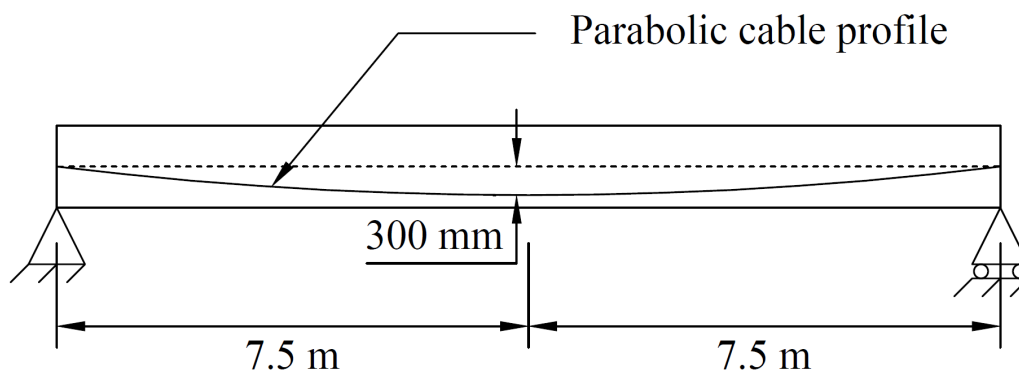


Figure 2.2 - Post-tensioning cable profile

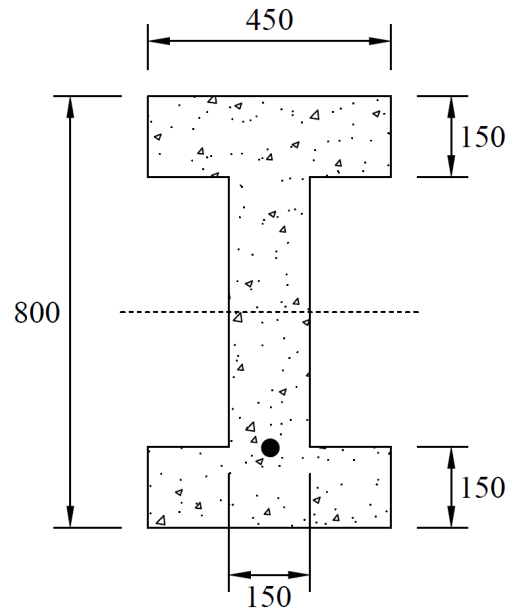


Figure 2.3 - Uncracked prestressed beam section

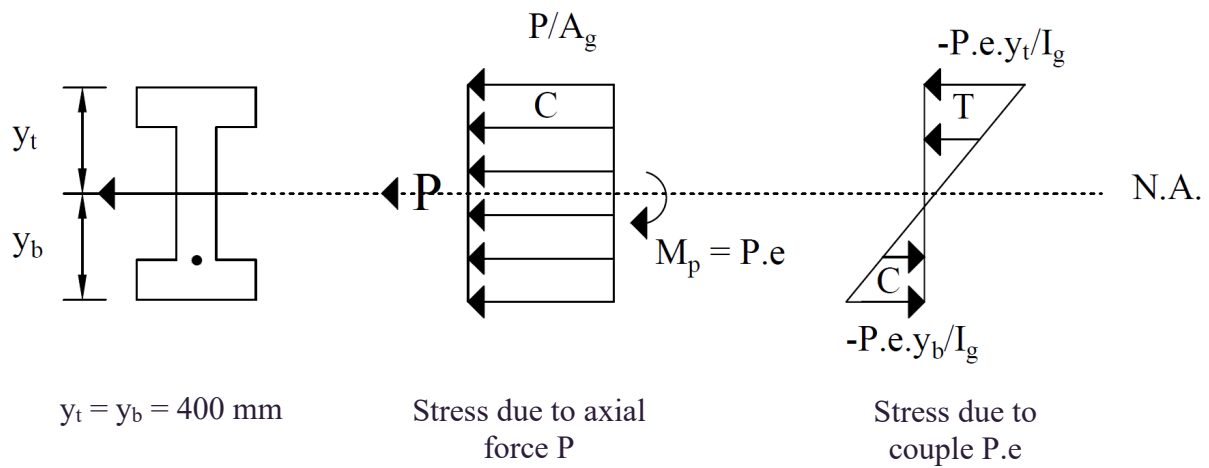


Figure 2.4 - Stress profiles of beam section

$$P = 80\% \times 184 \times 8 \text{ (number of strands)}$$

$$P = 1177.6 \text{ kN}$$

$$e = 300 \text{ mm}$$

1. Stresses in the beam section at midspan due to prestress acting alone:

At the top fibre of the beam:

$$\sigma_t = \frac{P}{A_g} - \frac{Pey_t}{I_g} \text{ (sign convention: compression +ve)}$$

$$\sigma_b = \frac{P}{A_g} + \frac{Pey_b}{I_g}$$

$$\sigma_t = \frac{1177.6 \times 10^3}{210,000} - \frac{1177.6 \times 10^3 \times 300 \times 400}{1.61 \times 10^{10}} = 5.6 - 8.8 = -3.2 \text{ MPa}$$

$$\sigma_b = \frac{1177.6 \times 10^3}{210,000} + \frac{1177.6 \times 10^3 \times 300 \times 400}{1.61 \times 10^{10}} = 5.6 + 8.8 = 14.4 \text{ MPa}$$

(note that the bottom of the beam is in compression)

2. Decompression moment at midspan:

To achieve decompression; i.e. zero bottom fibre stress, the bending moment M_{dec} must reduce the bottom fibre compressive stress of 14.4 MPa due to prestress, to zero. M_{dec} is therefore the bending moment which by itself would induce a bottom fibre tensile stress of 14.4 MPa.

$$M_{dec} = \frac{\sigma_{bp} I_y}{y_b} = \frac{14.4 \times 1.61 \times 10^{10}}{400} = 579.6 \text{ kNm}$$

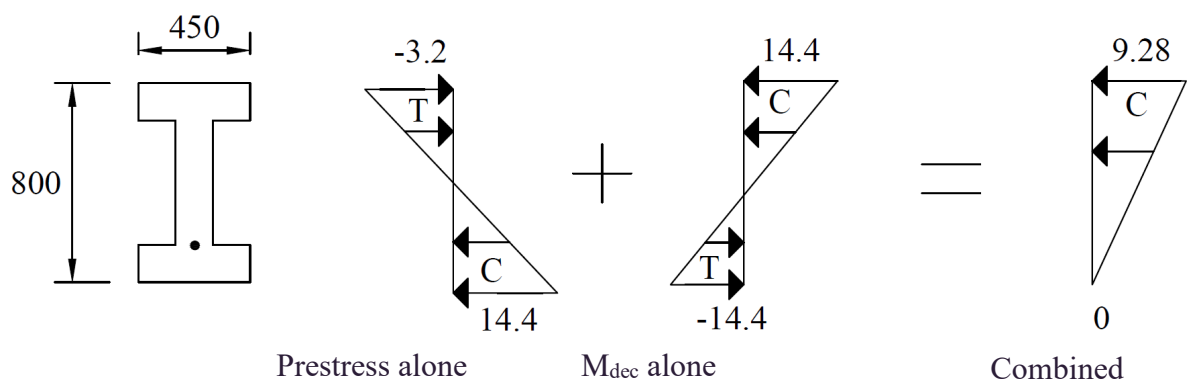


Figure 2.5 - Decompression moment at midspan

3. Stresses at mid-span due to prestress, self-weight and a uniform live load of 18 kN/m.

Moment due to applied load (M_{app}):

$$\text{self-weight} = 25 \text{ kN} / \text{m}^3 \times 210,000 \times 10^{-6} = 5.25 \text{ kN} / \text{m}$$

$$\text{live load} = 18 \text{ kNm}$$

$$\text{total applied load} = 18 + 5.25 = 23.25 \text{ kN} / \text{m}$$

$$\text{total applied moment, } M_{app} = \frac{23.25 \times 15^2}{8} = 653.9 \text{ kNm}$$

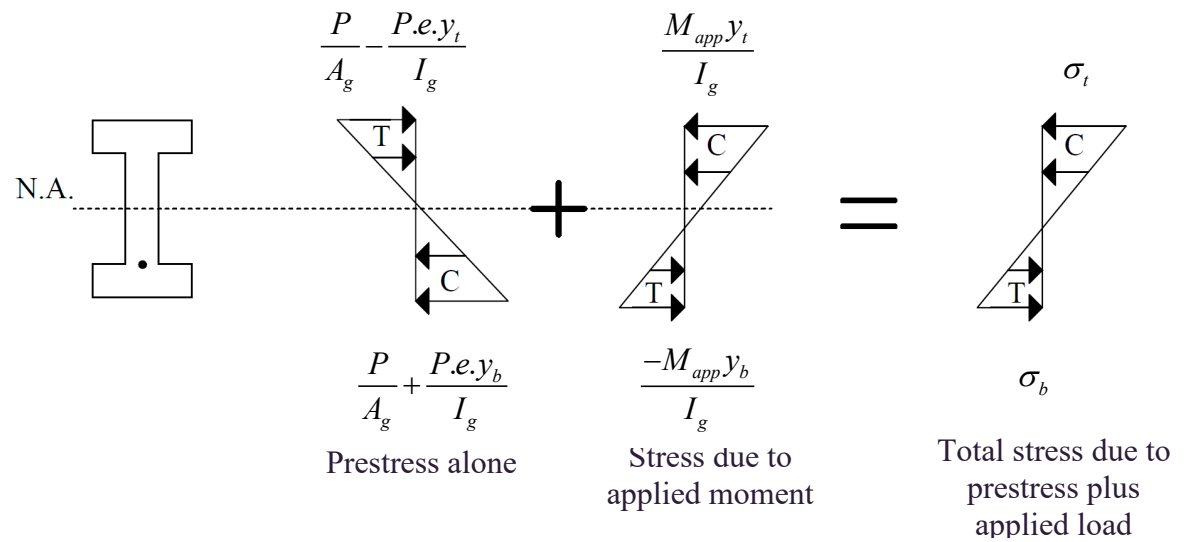


Figure 2.6 - Stresses at mid-span due to prestress, self-weight and uniform live load

At top fibre:

$$\sigma_t = \left(\frac{P}{A_g} - \frac{Pey_t}{I_g} \right) + \frac{M_{app} \cdot y_t}{I_g}$$

$$\sigma_t = -3.2 + \frac{653.9 \times 10^6 \times 400}{1.61 \times 10^{10}} = -3.2 + 16.2 = 13.0 \text{ MPa}$$

At bottom fibre:

$$\sigma_t = \left(\frac{P}{A_g} + \frac{Pey_b}{I_g} \right) - \frac{M_{app} \cdot y_b}{I_g}$$

$$\sigma_t = 14.4 - 16.2 = -1.8 \text{ MPa} \quad (\text{tension})$$



4. *Is the beam cracked under the above loading?*

Calculate cracking tensile stress (flexural strength) of concrete:

$$f_{cf}' = 0.6\sqrt{32} = 3.39 \text{ MPa}$$

$$\underbrace{-1.8 \text{ MPa}}_{\text{tensile stress due to applied load}} < \underbrace{3.39 \text{ MPa}}_{\text{flexural tensile stress of concrete}}$$

Therefore, beam is not cracked under the given load.